

08/17/06

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GWs from a migrating fragment

Detector response to binary waveform

$$h(t) = F_+ h_+(t) + F_x h_x(t) = \frac{\Theta}{D} M (\pi M f)^{2/3} \cos(\psi + \varphi)$$

 $F_{+,x}$ = antenna patterns

$$\Theta^2 = 4 [F_+^2 (1 + \cos^2 I) + 4 F_x^2 \cos^2 I] \in (0, 16)$$

$$M = \text{chirp mass} = \mu^{3/5} M_b^{2/5} \quad ; \quad M_b = M_1 + M_2 \quad ; \quad \mu = \frac{M_1 M_2}{M_b}$$

$$f = \text{GW frequency} = \Omega / \pi \quad ; \quad \Omega^2 = GM_b / R^3$$

$$= \frac{1}{2\pi} \frac{d\phi}{dt}$$

 φ = constant phaseFragment parameterseffective R_f

fragment forms at disk radius

$$R_f = (100 R_2) \underbrace{\frac{GM_b}{c^2}}_{\approx 450 R_2 \left(\frac{M_b}{3M_\odot}\right) \text{ km}}$$

$$\Omega_f^2 = \frac{GM_b}{R_f^3} = \left(\frac{c^3}{GM_b}\right)^2 \frac{1}{10^6 R_2^3} \approx 70 R_2^{-3/2} \left(\frac{M_b}{3M_\odot}\right)^{-1} \text{ rad/s}$$

$$\Rightarrow f_f \approx \underline{20} R_2^{-3/2} \left(\frac{M_b}{3M_\odot}\right)^{-1} \text{ Hz}$$

if the fragment has mass M_f and is supported by neutron degeneracy, then its radius is

$$r_f \sim 10 \text{ km} \left(\frac{M_f}{1.4 M_\odot}\right)^{-1/3} \approx 24 \left(\frac{M_f}{0.1 M_\odot}\right)^{-1/3} \text{ km}$$

tidal disruption at radius

$$R_t \approx 2 r_f \left(\frac{M_b}{M_f}\right)^{1/3} \approx 30 \left(\frac{M_\odot}{M_f}\right)^{2/3} \left(\frac{M_b}{3M_\odot}\right)^{1/3} \text{ km}$$

the frequency at R_t : $f_t \approx 1.1 \left(\frac{M_f}{M_b} \right) \text{ kHz}$

chirp mass: $M = \left(\frac{M_f}{M_b} \right)^{3/5} \left(1 - \frac{M_f}{M_b} \right)^{3/5} M_b$
(typically $\sim 1 M_b$)

Strain amplitude

$$h(t) = h_0(t) \cos(\phi + \psi) \quad ; \quad h_0 \sim \frac{M}{D} (\pi M f)^{2/3} \\ \equiv \frac{M}{D} (M \Omega)^{2/3}$$

$$h_0 \approx \frac{M}{D} \left[\frac{M}{M_b} \left(\frac{R}{R_g} \right)^{-3/2} \right]^{2/3} = \frac{M}{D} \frac{R_g}{R} \left(\frac{M}{M_b} \right)^{2/3} \\ \equiv \frac{G M}{c^2 D} \frac{G M_b}{c^2 R}$$

$$h_0 \sim 6.4 \times 10^{-24} \left(\frac{M}{M_b} \right)^{2/3} \left(\frac{D}{100 \text{ Mpc}} \right)^{-1} \left(\frac{f}{100 \text{ Hz}} \right)^{2/3}$$

for a band-pass filter with frequency $\sim f$ and noise $\sqrt{f S_h(f)} \sim 10^{-23}$, the S/N for the above fiducial parameters is ~ 1 . This is somewhat encouraging since a matched filter will provide a gain of $\sim \sqrt{\# \text{ cycles}}$.

For a matched filter, (see Flanagan & Hughes '98)

$$\left(\frac{S}{N} \right)^2 = 4 \int_0^\infty df \frac{|\tilde{h}(f)|^2}{S_h(f)} ;$$

S_h = noise spectrum s.t. $\sqrt{f S_h(f)}$ is the broadband strain noise ;

$$\tilde{h}(f) = \int_{-\infty}^{\infty} dt e^{i2\pi f t} h(t).$$

The stationary phase approximation gives

$$|\tilde{h}(f)|^2 = \frac{1}{4} h_0^2(t(f)) \ddot{f}^{-1}(f)$$

$$\Rightarrow \left(\frac{S}{N}\right)^2 = \int_0^{\infty} dt \ddot{f}^{-1}(f) \frac{h_0^2(t(f))}{S_n(f)}$$

$$h_n^2(f) = f S_n(f) \Rightarrow \left(\frac{S}{N}\right)^2 = \int_0^{\infty} d \ln f \left(\frac{f}{f}\right) \frac{h_0^2(t(f))}{h_n^2(f)}$$

$$dN_{\text{cyc}} = \frac{1}{2\pi} d\psi = dt f = df \frac{f}{\dot{f}} = d \ln f \frac{f^2}{\dot{f}}$$

$$\Rightarrow \frac{dN_{\text{cyc}}}{d \ln f} = \frac{f^2}{\dot{f}}$$

$$\Rightarrow \boxed{\left(\frac{S}{N}\right)^2 = \int_0^{\infty} d \ln f \frac{dN_{\text{cyc}}}{d \ln f} \frac{h_0^2(f)}{h_n^2(f)}}$$

$\Rightarrow \sqrt{\frac{dN_{\text{cyc}}}{d \ln f}}$ is an effective signal gain per log frequency

In practice, the integral over the inspiral phase ranges from $\underline{f_f}$ to $\underline{f_t}$.

Fragment migration

The dynamics of migration are contained in $\dot{f}(t)$. The faster f sweeps through a frequency the smaller the number of cycles and the lower the gain.

For pure GW driven inspiral:

$$\frac{f^2}{\dot{f}} = \frac{dN_{\text{cyc}}}{d \ln f} = \frac{5}{96\pi} (\pi M f)^{-5/3} \approx 800 \left(\frac{M}{M_\odot} \frac{f}{100 \text{ Hz}} \right)^{-5/3}$$

$$\begin{aligned} \frac{dN_{\text{cyc}}}{d \ln f} h_0^2(f) &= \frac{\Theta^2}{D^2} \frac{5}{96\pi} (\pi M f)^{-5/3} \cdot M^2 (\pi M f)^{4/3} \\ &= \frac{\Theta^2}{D^2} \frac{5}{96\pi} M^2 (\pi M f)^{-1/3} \end{aligned}$$

$$\textcircled{X} \Rightarrow \left(\frac{S}{N} \right)^2 = \frac{\Theta^2}{D^2} \frac{5}{96\pi} M^2 (\pi M)^{-1/3} \int_{f_f}^{f_t} df f^{-1/3} [f S_h(f)]^{-1}.$$

Approximating $(f S_h(f))^{1/2}$ as a constant h_n over (f_f, f_t) , we find

$$\begin{aligned} \left(\frac{S}{N} \right)^2 &= \frac{\Theta^2}{D^2} \frac{5}{96\pi} M^2 (\pi M)^{-1/3} h_n^{-2} \times 3 [f_f^{-1/3} - f_t^{-1/3}] \\ &= \Theta^2 \left(\frac{M}{D} \right)^2 \frac{5}{32\pi} (\pi M f_f)^{-1/3} h_n^{-2} \left[1 - \left(\frac{f_f}{f_t} \right)^{1/3} \right] \end{aligned}$$

$$\frac{S}{N} = \Theta \left(\frac{S}{32\pi} \right)^{1/2} \frac{M}{D} (\pi M f_f)^{-1/6} h_n^{-1} \left[1 - \left(\frac{f_f}{f_t} \right)^{1/3} \right]^{1/2};$$

or alternatively,

$$\frac{S}{N} = \left[3 \frac{dN_{\text{cyc}}}{d \ln f} (f_f) \right]^{1/2} [h_0(f_f)/h_n] \left[1 - \left(\frac{f_f}{f_t} \right)^{1/3} \right]^{1/2}.$$

The matched-filter signal-to-noise for GW inspiral is largely determined by the lowest frequencies, where the largest number of cycles is accumulated.

Quantitatively, we have

$$\frac{S}{N} \approx \underline{30} \left(\frac{D}{100 \text{ Mpc}} \right)^{-1} \left(\frac{M}{M_\odot} \right)^{5/6} \left(\frac{P_{ff}}{100 \text{ Hz}} \right)^{-1/6} h_{23}^{-1} \underbrace{\left[1 - \left(\frac{P_{ff}}{f_t} \right)^{1/3} \right]^{1/2}}_{\sim 1};$$

(x 10)

$$h_{23} = h_n / 10^{-23}.$$

Now suppose the primary driver of migration is the viscous torque in the disk. The fragment opens a gap in the disk if

$$M_f / M_\bullet \gtrsim \alpha^{1/2} \left(\frac{H}{R} \right)^2 \equiv \alpha^{1/2} \eta^2 = 0.01 \left(\frac{\alpha}{10^{-2}} \right)^{1/2} \left(\frac{\eta^2}{0.1} \right).$$

↳ BH mass

Gap formation seems likely when $M_f \gtrsim 0.1 M_\odot$. This corresponds to Type II migration, where the migration rate is set by the viscous time in the disk:

$$\left| \frac{\dot{R}}{R} \right| \sim \frac{2}{R^2} = \frac{\alpha c_s H}{R^2}; \quad c_s = H \Omega \Rightarrow \underline{\underline{\alpha \Omega \eta^2}} = \underline{\underline{\pi f \alpha \eta^2}};$$

$$\frac{\dot{f}}{f} = -\frac{3}{2} \frac{\dot{R}}{R} = \frac{3\pi}{2} f \alpha \eta^2$$

$$\Rightarrow \frac{f^2}{\dot{f}} \approx \frac{2}{3\pi} \underline{\underline{(\alpha \eta^2)^{-1}}} \sim \underline{\underline{(\alpha \eta^2)^{-1}}} \sim \underline{10^3} \left(\frac{\alpha}{0.01} \frac{\eta^2}{0.1} \right)^{-1}.$$

In general, we sum the orbital torques from GWs and the disk:

$$\begin{aligned} \left(\frac{\dot{S}}{S}\right)_{\text{total}} &= \left(\frac{\dot{S}}{S}\right)_{\text{gw}} + \left(\frac{\dot{S}}{S}\right)_d = \frac{1}{2} \left(\frac{\dot{R}}{R}\right)_{\text{tot}} = \frac{1}{2} \left[\left(\frac{\dot{R}}{R}\right)_{\text{gw}} + \left(\frac{\dot{R}}{R}\right)_d \right] \\ &= -\frac{1}{3} \left[\left(\frac{\dot{f}}{f}\right)_{\text{gw}} + \left(\frac{\dot{f}}{f}\right)_d \right] = -\frac{1}{3} \left(\frac{\dot{f}}{f}\right)_{\text{tot}}. \end{aligned}$$

The GW and disk contributions are equal when

$$\left(\frac{\dot{f}}{f}\right)_{\text{gw}} = \left(\frac{\dot{f}}{f}\right)_d \equiv \left(\frac{\dot{f}}{f^2}\right)_{\text{gw}} = \left(\frac{\dot{f}}{f^2}\right)_d$$

$$\Rightarrow \frac{96\pi}{5} (M f_{\text{eq}})^{5/3} = \alpha \eta^2$$

$$\Rightarrow f_{\text{eq}} = \frac{1}{\pi M} \left(\frac{5\alpha \eta^2}{96\pi} \right)^{3/5} \approx 90 \left(\frac{M}{M_{\odot}} \right)^{-1} \left[\frac{\alpha}{0.01} \frac{\eta^2}{0.1} \right]^{3/5} \text{ Hz};$$

$$f > f_{\text{eq}} \Rightarrow \text{GWs dominate}.$$

It may be that disk torques dominate only for the initial migration, but this may still cut the GW S/N substantially.

If $f_f < f_{\text{eq}} < f_+$, then the matched-filter S/N is given by

$$\left(\frac{S}{N}\right)^2 \approx \left\{ \int_{f_f}^{f_{\text{eq}}} df \left(\frac{dW_{\text{acc}}}{df} \right)_d + \int_{f_{\text{eq}}}^{f_+} df \left(\frac{dW_{\text{acc}}}{df} \right)_{\text{gw}} \right\} \times \left(h_{\text{oz}}^2(f) / h_n^2(f) \right).$$

If we again fix h_n , the the disk and GW pieces are:

disk

$$\begin{aligned} \left(\frac{S}{N}\right)_d^2 &= (\alpha \eta^2)^{-1} \left(\frac{G}{D}\right)^2 M^2 (R M)^{4/3} h_n^{-2} \int_{f_t}^{f_{eq}} df f^{-1} f^{4/3} \\ &= (\alpha \eta^2)^{-1} \cdot \frac{3}{4} f_{eq}^{4/3} \left[1 - \left(\frac{f_t}{f_{eq}}\right)^{4/3}\right] \\ &\quad \times h_n^2 = \frac{3}{4} h_0^2 (f_{eq}) \left[1 - \left(\frac{f_t}{f_{eq}}\right)^{4/3}\right]; \end{aligned}$$

$$\left(\frac{S}{N}\right)_{gw}^2 = \left[3 \left(\frac{dN_{gw}}{d \ln f}\right)_{gw} (f_{eq})\right] h_0^2 (f_{eq}) h_n^{-2} \left[1 - \left(\frac{f_t}{f_{eq}}\right)^{4/3}\right]$$

$$\Rightarrow \left(\frac{S}{N}\right)_d^2 + \left(\frac{S}{N}\right)_{gw}^2 =$$

$$\begin{aligned} &\left\{ \frac{3}{4} \left(\frac{dN_{gw}}{d \ln f}\right)_d (f_{eq}) \left[1 - \left(\frac{f_t}{f_{eq}}\right)^{4/3}\right] \right. \\ &\quad \left. + 3 \left(\frac{dN_{gw}}{d \ln f}\right)_{gw} (f_{eq}) \left[1 - \left(\frac{f_t}{f_{eq}}\right)^{4/3}\right] \right\} \\ &\quad \times \frac{h_0^2 (f_{eq})}{h_n^2} . \end{aligned}$$

Quantitatively,

$$\begin{aligned} \left(\frac{S}{N}\right)^2 &= \left\{ 750 \left[1 - \left(\frac{f_t}{f_{eq}}\right)^{4/3}\right] + 2400 \left(\frac{M}{M_\odot} \frac{f_{eq}}{100 \text{ Hz}}\right)^{-5/3} \left[1 - \left(\frac{f_t}{f_{eq}}\right)^{4/3}\right] \right\} \\ &\quad \times G^2 \left\{ 0.64 h_{23}^{-1} \left(\frac{M}{M_\odot}\right)^{5/3} \left(\frac{D}{100 \text{ Mpc}}\right)^{-1} \left(\frac{f_{eq}}{100 \text{ Hz}}\right)^{2/3} \right\}^2 . \end{aligned}$$

If $f_{eq} \rightarrow f_f$, we recover the GW result. If, e.g., M is small, or $\alpha\gamma^2$ is large, we may have $f_{eq} \rightarrow f_t$ and obtain a gain of $\sqrt{750} \approx 30$ and a net signal-to-noise of ≈ 20 for fiducial parameter values.